

Project Documentation

Documentation Format — Cosmetic Insights (Tableau BI Dashboard)

This project is a Tableau-based Business Intelligence dashboard, not a full-stack web application. The standard documentation sections below have been adapted from the program's Project Documentation format to fit a BI/analytics deliverable.

1. Introduction

Project Title: Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

Team Members and Roles:

- Maheswar Reddy Kathithatagari — Team Lead
- Vijay Kumar Gogula — Data Upload & Preprocessing, Skin-Type Segmentation
- Yagna Mekala — Data Preprocessing, Skin-Type Coverage Analysis
- Joshna Vutupalli — Brand & Pricing Analysis, Dashboard Interactivity
- Guggulla Sree Pranavi — User Access & Roles, Reporting/Export

2. Project Overview

Purpose:

To give cosmetics brand managers, analysts, and leadership a single, connected view of brand, price, rank, and skin-type data so that pricing, marketing, and product decisions can be made quickly and with evidence, instead of by manually reconciling scattered spreadsheets.

Features:

- Brand and category performance comparison by price and rank
- Skin-type coverage view (Dry, Oily, Combination, Normal, Sensitive) across brands and categories
- Interactive filters and drill-down from category summaries to individual products
- Role-based views for Analysts vs. Leadership
- Exportable summary reports for review meetings

3. Architecture

Data Layer (in place of “Database”):

The cosmetics dataset (brand, category, price, rank, skin type) is stored as a cleaned CSV/Excel extract, with an optional MySQL store for larger volumes.

Processing Layer (in place of “Backend”):

Data cleaning and structuring is handled in Tableau Prep (or Python/Pandas for heavier preprocessing) before being loaded into Tableau's data model, with calculated fields and LOD expressions applied for analysis.

Presentation Layer (in place of “Frontend”):

The interactive dashboard is built in Tableau Desktop and published to Tableau Public/Server/Cloud, where users view and filter it through a web browser.

4. Setup Instructions

Prerequisites:

- Tableau Desktop or Tableau Public
- The cosmetics dataset file (CSV/Excel)
- Tableau Server/Cloud account for publishing (optional for local viewing)

Installation:

- Install Tableau Desktop/Public and open the project workbook (.twbx)
- Verify the data source connection points to the cosmetics dataset file
- Refresh the data source extract to load the latest data

5. Folder Structure

Workbook (in place of “Client”):

Contains the Tableau worksheets, dashboards, and the story that make up the user-facing views.

Data Sources (in place of “Server”):

Contains the raw and cleaned dataset files, plus any Tableau Prep flows used to produce the cleaned extract.

6. Running the Application

- Open the .twbx workbook file in Tableau Desktop, or
- Open the published dashboard link in a browser via Tableau Public/Server/Cloud

7. API Documentation

Not applicable — Cosmetic Insights is a self-contained Tableau dashboard with no custom backend API. All data access happens through Tableau's native data connections.

8. Authentication

Access is managed through Tableau Server/Cloud sign-in. Role-based permissions restrict Analysts to the detailed working view and Leadership to a read-only summary view.

9. User Interface

[Insert screenshots of the Brand & Pricing view, Skin-Type Coverage view, and the summary Story here.]

10. Testing

Covered in the Project Development Phase: performance testing against the Tableau workbook (data rendering, preprocessing, filters, calculated fields, dashboard and story design) and User Acceptance Testing with 6 test cases, all passing.

11. Screenshots or Demo

[Insert dashboard screenshots or a link to the published Tableau Public/Cloud demo here.]

12. Known Issues

- Skin-type heatmap legend overlapped the chart on smaller screens — resolved by resizing the legend container
- Price filter reset previously cleared the skin-type filter as well — resolved by scoping the reset action

13. Future Enhancements

- Predictive trend indicators for emerging categories
- Automated alerts for sudden shifts in consumer preference
- Competitor benchmarking view
- Sentiment analysis of customer reviews
- Mobile-friendly version of the dashboard